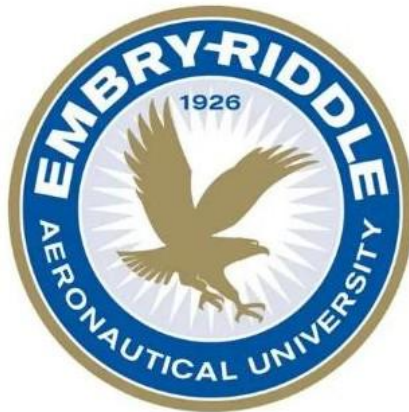


SRV02 Position Control



Professor: Bernardo Restrepo

Graduate Assistant: Jared Getz

Author: Anuranan Bharadwaj

AE 443: Experimental Dynamics and Control Laboratory

Department of Aerospace Engineering

Embry-Riddle Aeronautical University

Daytona Beach, FL, 32114

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Procedure

A. Simulation Procedure

The experiment started by first examining the closed-loop behavior of the SRV02 system with the PV controller in simulation. The provided Simulink model was used in this step. To import the system parameters and initialize the model, the setup script was run, while the controller blocks were updated using the proportional and velocity gains determined in the pre-lab. The signal generator was then set up to generate a ramp-type input from a triangle reference signal. The desired ramp reference was applied to the system by adjusting the amplitude and frequency parameters in accordance with the lab instructions. After running the simulation, the scopes in the Simulink file were used to track the load shaft position and input motor voltage. Finally, the results were exported as MATLAB figures for plotting.

B. Experimental Procedure

The same setup was used on actual SRV02 hardware once the controller behavior was verified in simulation. To establish the interaction between Simulink and the hardware system, the experimental setup script was executed. The model was updated using the proportional and velocity controller gains from the pre-lab calculations. The signal generator was set up to generate a triangle reference input that matched the simulation's amplitude and frequency. In order to see the hardware response in real time, the system was then executed using Simulink's Monitor & Tune function. The scopes were used to record the motor input voltage and load shaft position. The measured ramp response was displayed in a MATLAB figure, and the response plot was used to calculate the steady-state error.

C. Hypothesis

The experiment's hypothesis is that the steady-state error seen during ramp tracking can be eliminated by adding an integral term to the existing PV controller. The system, when equipped with only proportional and derivative controllers, behaves as a Type-1 system and generates a constant steady-state error. The PID controller constantly modifies the control input by accumulating the error over time through the introduction of an integral component. Ultimately, this drives the steady-state error to zero, enabling the system output to more precisely track the reference.

D. Independent and Dependent Variables

The independent variables in the controller design are the control gains, i.e., the proportional gain k_p , velocity gain k_v , and integral gain k_i . These variables have a

major effect on the system's output and are chosen during controller design based on the desired goal that needs to be achieved by the system. The final system outputs or the measured responses, such as the motor input voltage, the steady-state error, and the load shaft position, are the dependent variables. These parameters illustrate the performance of the controller and how close the system's output is to the reference input.

E. Assumptions

During the controller design and experiment, a number of assumptions were made. Initially, it was considered that the controller design's mathematical model accurately represented the SRV02 system. This presumption makes it possible to apply the model's gains to the real system. Additionally, it was believed that nonlinear factors like friction, sensor noise, and actuator saturation were negligible and would not have a substantial impact on the system response within the experiment's operating range. Furthermore, it was anticipated that the motor and amplifier, among other hardware elements, would function within their linear bounds throughout the experiment.

F. Experimental Procedure

Simulation:

The integral control term was added to the Simulink model, resulting in a PIV controller. The controller block was updated with the integral gain computed in the pre-lab. The system was given the identical triangle reference input that was used in the earlier ramp experiment. After the simulation was run, the scopes were used to monitor the load shaft position and motor input voltage. These results were exported to MATLAB to confirm whether the integral term was doing its job in eliminating the steady-state error.

Implementation:

The identical configuration was used on the SRV02 hardware system after the controller's performance was confirmed in simulation. The controller was run on the physical system using the Simulink model in the Monitor & Tune mode. The scopes were used to track the motor voltage signals and load shaft location. Plots of the position response and input voltage were created by exporting the response data to MATLAB. To assess the performance of the PIV controller, the steady-state error was calculated from experimental data and compared with the simulation values.

Results

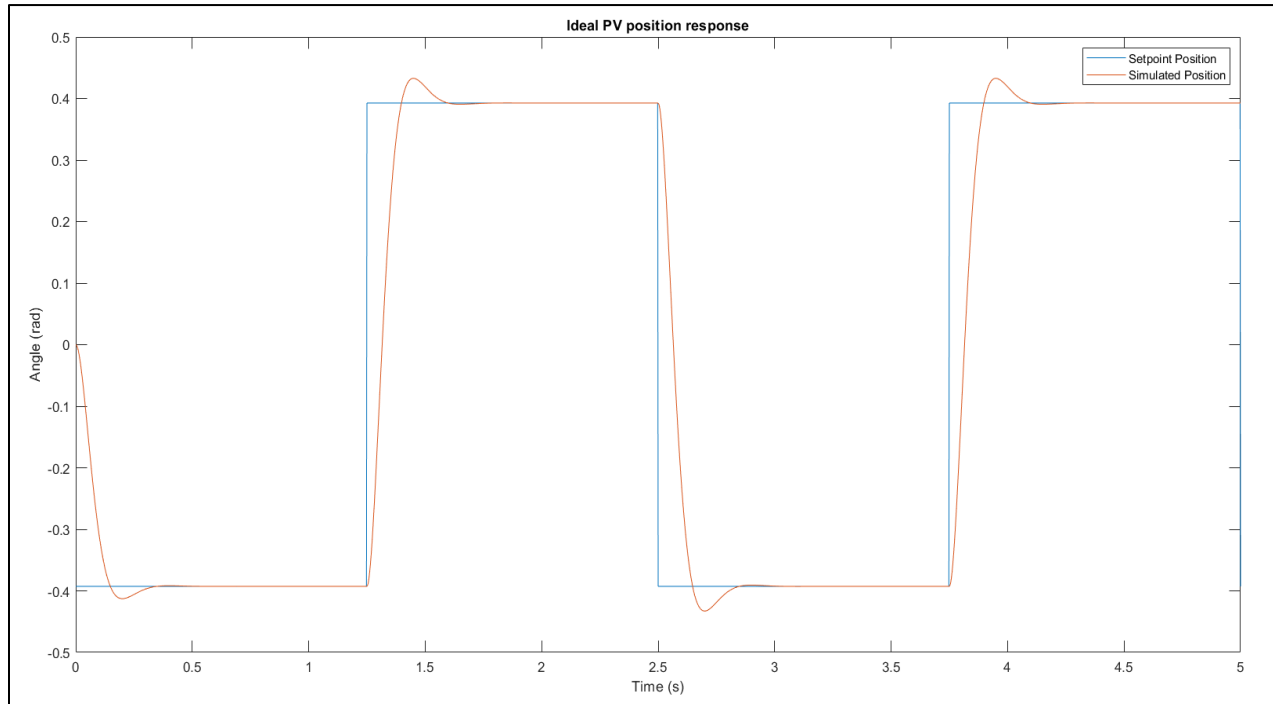


Figure 1: Simulated Step Response

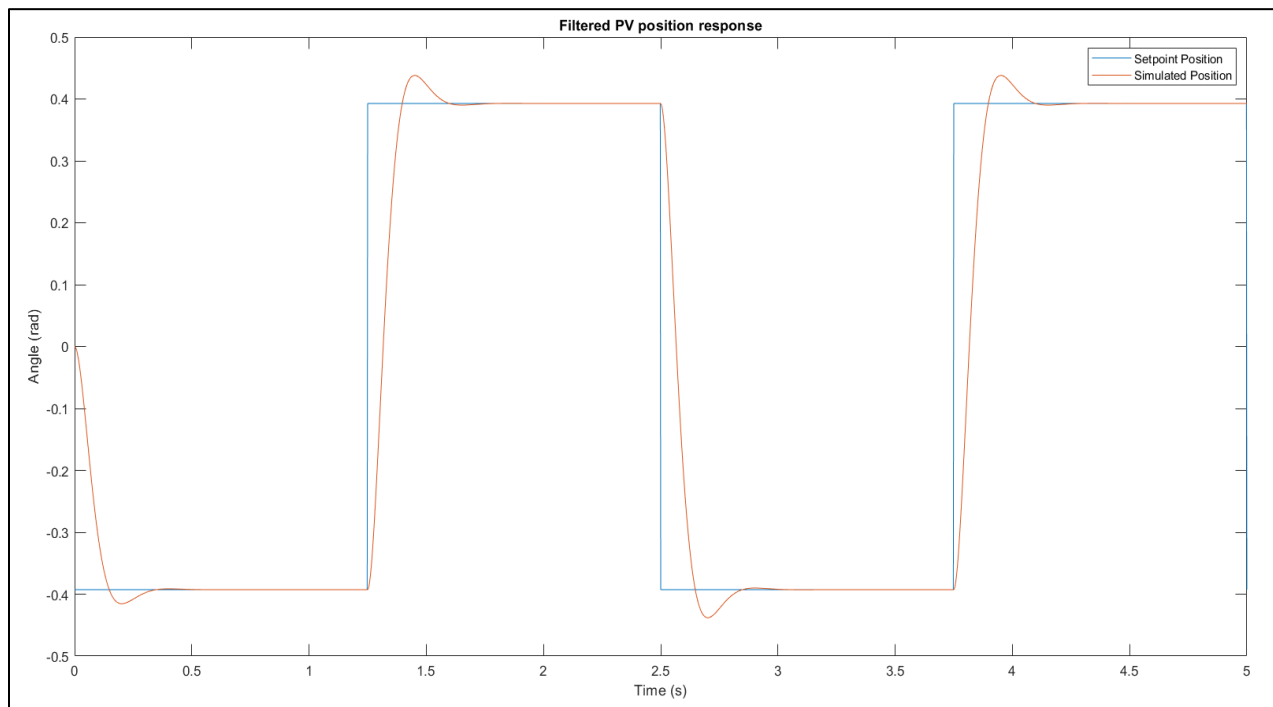


Figure 2: Filtered PV Response

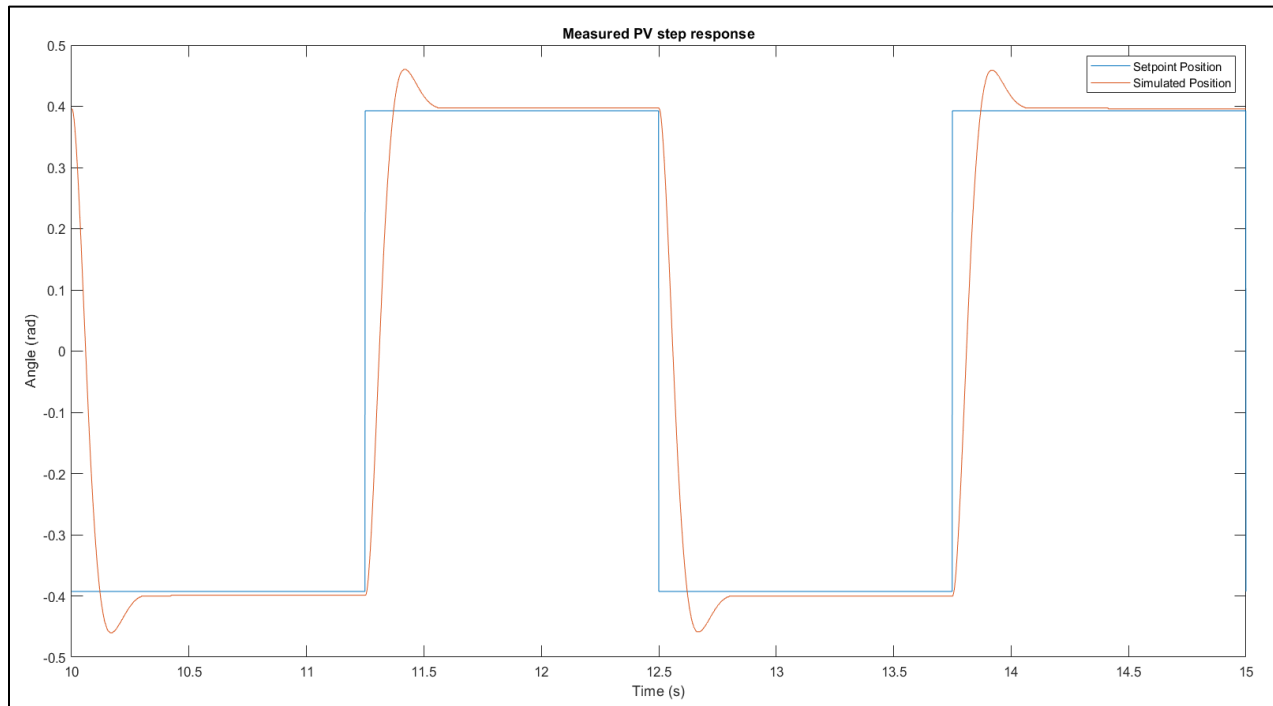


Figure 3: Step Response of Implemented PV Controller

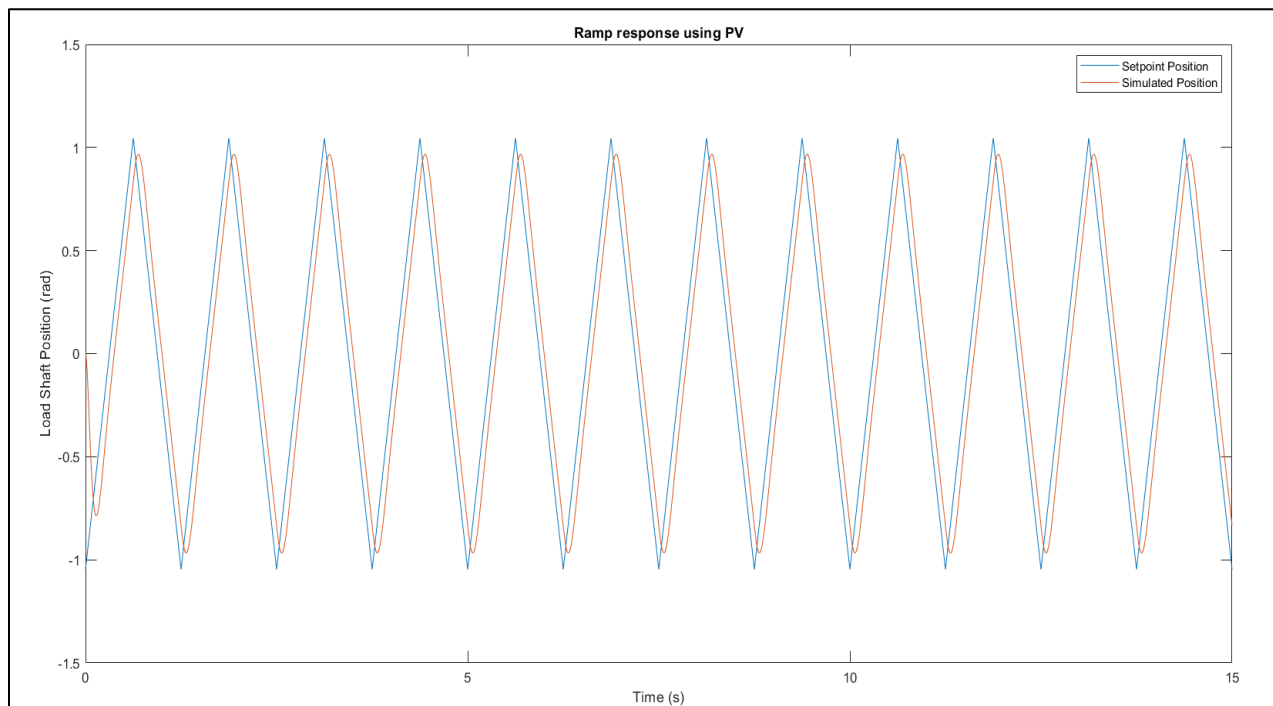


Figure 4: Simulated PV Controller with Ramp Input

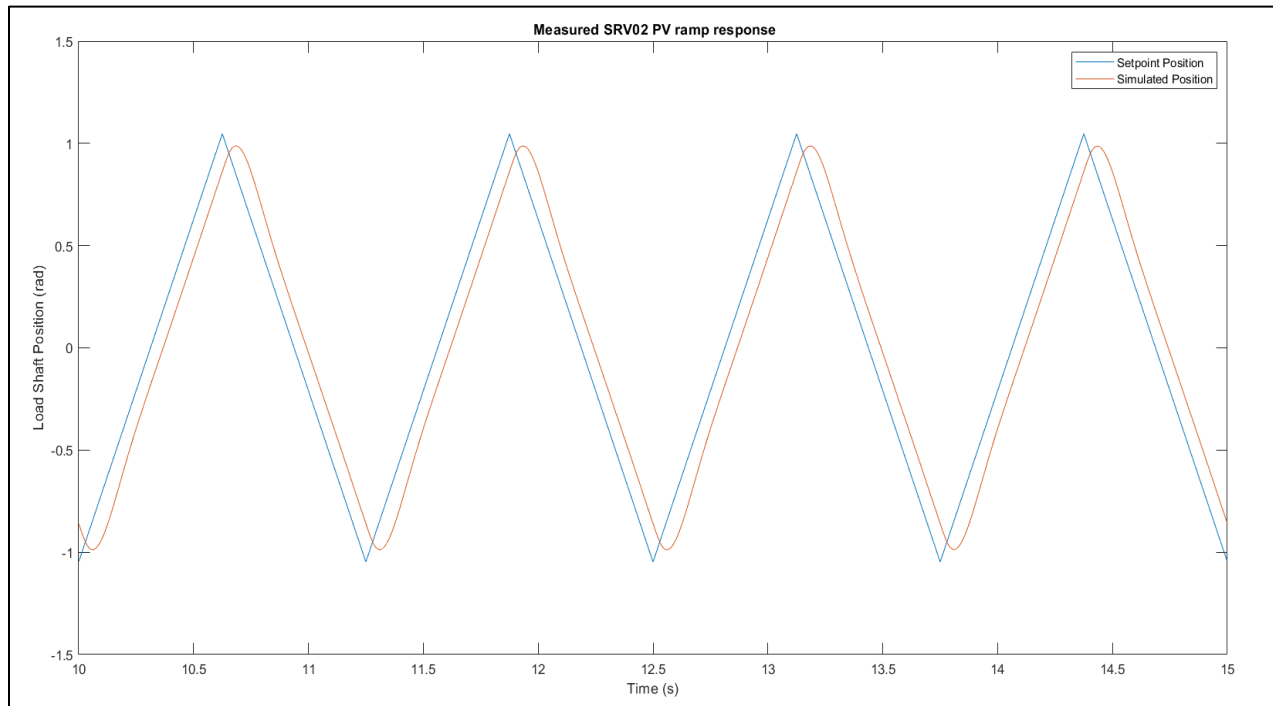


Figure 5: Ramp Response of Implemented PV Controller

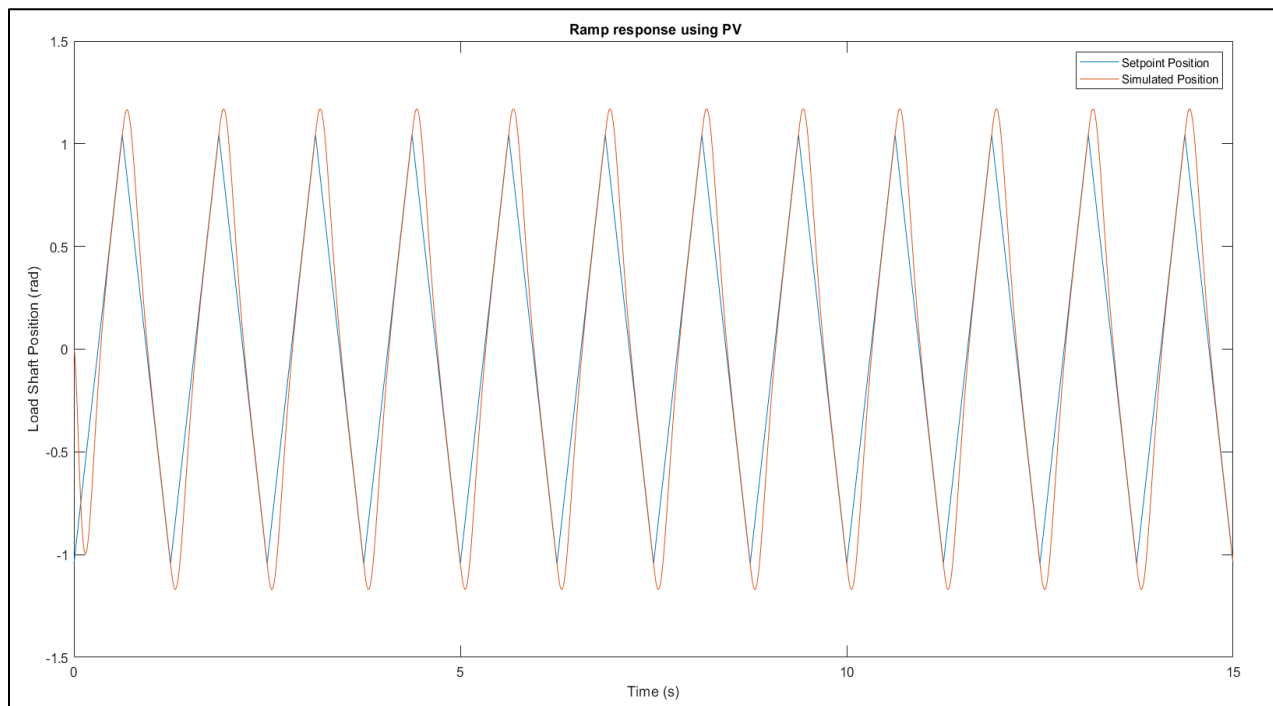


Figure 6: Simulated Controller with ramp Input

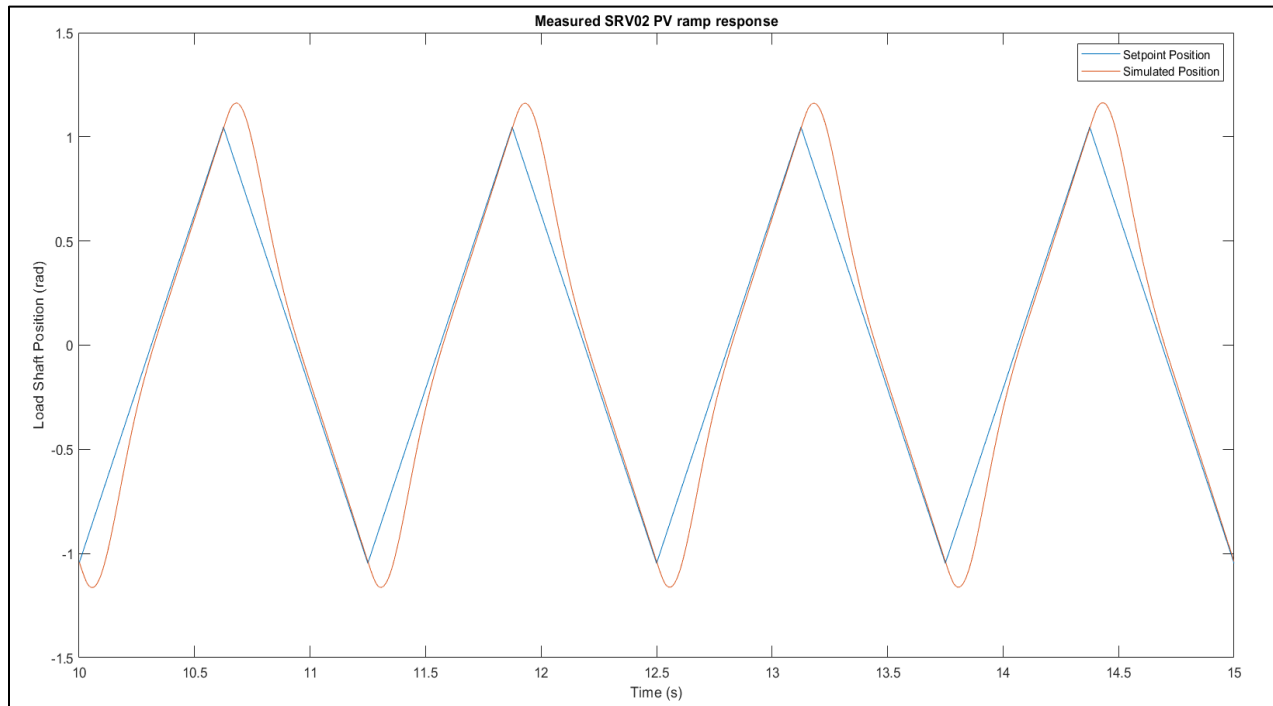


Figure 7: Implemented Controller with Ramp Input

Table 1: Summary of results for the SRV02 Position Control laboratory

Section	Description	Symbol	Value	Unit
1(a)	Pre-Lab: Model Parameters Open-Loop Steady-State Gain	K	1.53	rad/(V·s)
	Open-Loop Time Constant	τ	0.0254	s
1(b)	Pre-Lab: PV Gain Design Proportional gain	k_p	7.82	V/rad
	Velocity gain	k_v	-0.157	V·s/rad
2	Pre-Lab: Control Gain Limits	For 20deg case $k_{p,max}$	28.7	V/rad

	Maximum proportional gain	For 45deg case $k_{p,max}$	12.7	V·s/rad
3	Pre-Lab: Ramp Steady-State Error Steady-state error using PV	e_{ss}	0.214	rad
4	Pre-Lab: Integral Gain Design Integral gain	For t=1s case k_i For t=2s case k_i	38.9 19.5	V/(rad·s)
I.	Step Response Simulation Peak time Percent overshoot Steady-state error	t_p PO e_{ss}	0.194 5 0	s % rad
II.	Filtered Step Response Using PV Peak time Percent overshoot Stead-state error	t_p PO e_{ss}	0.204 5.7 0.002	s % Rad
III.	Step Response Implementation Peak time Percent overshoot Stead-state error	t_p PO e_{ss}	0.17 7.86 0.004	s % rad
IV.	Ramp Response Simulation with PV Stead-state error	e_{ss}	0.213	rad
V.	Ramp Response Implementation with PV	e_{ss}	0.186	rad

	Steady-state error			
VI.	Ramp Response Simulation with no steady-state error Steady-state error	e_{ss}	0.005	rad
VII.	Ramp Response Implementation with no steady-state error Steady-state error	e_{ss}	0.007	rad

Calculations for Pre-Lab:

- (a) In order to get vital parameters like gain design, control gain limits, steady-state error, etc of the position controller, the required transient response must first be converted into the corresponding second-order system parameters. The desired response has a peak time (t_p) of 0.20s and percent overshoot (PO) of 5.0 %. Nominal parameters of the SRV02 model, like $K = 1.53 \text{ rad}/(\text{V} \cdot \text{s})$ and $\tau = 0.0254 \text{ s}$.

(b) Damping Ratio is calculated from the percent overshoot:

$$\ln(PO) = -\frac{\pi\zeta}{\sqrt{1-\xi^2}}$$

$$\ln(0.05) = -\frac{\pi\zeta}{\sqrt{1-\xi^2}}$$

Solving for ζ gives,

$$\zeta = 0.69$$

Next, the natural frequency is obtained:

$$\omega_n = \frac{\pi}{t_p \sqrt{1-\zeta^2}}$$

$$\omega_n = \frac{\pi}{0.20 \sqrt{1-0.69^2}} = 21.7 \text{ rad/s}$$

These quantities enables to determine proportional and velocity gain:

$$k_p = \frac{\tau \omega_n^2}{K}$$

$$k_p = \frac{0.0254(21.7)^2}{1.53} = 7.82 \text{ V/rad}$$

$$k_v = \frac{2\zeta\omega_n\tau - 1}{K}$$

$$k_v = \frac{2(0.69)(21.7)(0.0254) - 1}{1.53} = -0.157 \text{ V} \cdot \text{s/rad}$$

2. With the amplifier limit $V_{max} = 10 \text{ V}$, the maximum allowable proportional gain depends on the magnitude of the reference step input R_0 . The following equation is used to calculate the maximum proportional gains:

$$k_{p,max} = \frac{V_{max}}{R_0}$$

Step input: $20^\circ/0.349 \text{ rad}$

$$k_{p,max} = \frac{10}{0.349} = 28.7 \text{ V/rad}$$

Step input: $45^\circ/0.785 \text{ rad}$

$$k_{p,max} = \frac{10}{0.785} = 12.7 \text{ V} \cdot \text{s/rad}$$

3. Since the system is Type-1 using PV control, a ramp input produces a non-zero state error that can be calculated with the following equation:

$$R = \frac{R_0}{s^2}$$

$$e_{ss} = \frac{R(1 + Kk_v)}{Kk_v} = \frac{3.36(1 + 1.53(-0.157))}{1.53(7.82)} = 0.214 \text{ rad}$$

4. To remove or reduce the steady-state error, an integral controller is added to the system, making it a PIV controller.

$$V_{max} = k_p e_{ss} + k_i (e_{ss} t)$$

$$k_i = \frac{V_{max} - k_p e_{ss}}{e_{ss} t}$$

For $t = 1 \text{ s}$:

$$k_i = \frac{10 - (7.82)(0.214)}{(0.214)(1)} = 38.9 \text{ V}/(\text{rad} \cdot \text{s})$$

For $t = 2 \text{ s}$:

$$k_i = \frac{10 - (7.82)(0.214)}{(0.214)(2)} = 19.5 \text{ V}/(\text{rad} \cdot \text{s})$$

Analysis

The following section illustrates the calculations performed to determine key parameters such as peak time, steady-state error, and percentage overshoot.

I. Step response with PV controller

$$t_p = t_{max} - t_0 = 1.448 - 1.254 = 0.194 \text{ s}$$

$$PO = \frac{100(y_{max} - R_0)}{R_0} = \frac{100(0.84 - 0.8)}{0.8} = 5 \%$$

$$e_{ss} = 0.393 - 0.393 = 0 \text{ rad}$$

II. Step response with PV controller using low-pass filter

$$t_p = t_{max} - t_0 = 1.448 - 1.244 = 0.204 \text{ s}$$

$$PO = \frac{100(y_{max} - R_0)}{R_0} = \frac{100(0.853 - 0.807)}{0.807} = 5.7 \%$$

$$e_{ss} = 0.393 - 0.391 = 0.002 \text{ rad}$$

III. Step response with PV controller

$$t_p = t_{max} - t_0 = 11.42 - 11.25 = 0.17 \text{ s}$$

$$PO = \frac{100(y_{max} - R_0)}{R_0} = \frac{100(0.919 - 0.852)}{0.852} = 7.86 \%$$

$$e_{ss} = 0.397 - 0.393 = 0.004 \text{ rad}$$

IV. Simulated PV controller with ramp input

$$e_{ss} = 0.896 - 0.683 = 0.213 \text{ rad}$$

- V. Ramp response of the implemented PV controller

$$e_{ss} = 0.481 - 0.295 = 0.186 \text{ rad}$$

- VI. Simulated controller with ramp input

$$e_{ss} = 0.729 - 0.724 = 0.005 \text{ rad}$$

- VII. Implemented controller with ramp input

$$e_{ss} = 0.960 - 0.953 = 0.007 \text{ rad}$$

Conclusion

A. Step response simulation with PV controller

The simulated step response of the PV controller illustrated the system's ability to rapidly and consistently reach the desired reference point. The system demonstrated a fast response, which can be verified by the measured peak time of approximately 0.194 seconds. The controller effectively managed the reaction without inducing significant oscillations, as indicated by the calculated percent overshoot of 5%. Furthermore, the output position ultimately aligned precisely with the intended reference position in the simulation, as demonstrated by the steady-state error of 0 rad. Overall, the findings show that the PV controller gave the step input in the simulated system, steady and precise position control.

B. Step response simulation with PV controller using a low-pass filter

The system's overall behavior was consistent with just minor performance changes when the low-pass filter was added to the PV controller simulation. The filter caused the peak time to slightly increase to about 0.204 s, indicating a slower response. The measured percentage overshoot was 5.7%, which is still within a reasonable range. The steady-state error was 0.002 rad, which is practically negligible. The results indicate that the addition of the low-pass filter improved tracking performance and smoothed the system's response.

C. Step response with the implemented PV controller

When the PV controller was applied to the actual SRV02 system, the response was closely the same as the simulated scenario, although minor discrepancies were observed due to real-world system influences. The system maintained a rapid reaction to the reference input; a measured peak time of 0.17 seconds. The slight increase in percentage overshoot to 7.86% could be attributed to unmodeled dynamics, noise, or hardware limitations. The implemented controller demonstrated

precise tracking of the desired position, as reflected in a steady-state error of 0.004 rad. These observations confirm the effective performance of the simulated controller within the actual system.

D. Ramp response with the implemented PV controller

The implemented system generated a steady-state error of roughly 0.186 rad, whereas the simulated system had a steady-state error of roughly 0.213 rad. This effect is expected as steady-state inaccuracy in tracking a ramp input cannot be eliminated by a PV controller alone. These findings support the theoretical prediction that, when using simply proportional and velocity feedback, a ramp input leads to steady-state error.

E. Both simulated and implemented modified controllers

The results much improved when the controller added an integral gain, as it reduced the steady-state error. The implemented system generated a steady-state error of 0.007 rad, whereas the simulation's steady-state error dropped to about 0.005 rad. Since both values are close to zero, it shows that the redesigned controller was successful in tracking the ramp input. The minor discrepancies observed between simulation and execution are likely attributable to real-world factors, including sensor noise, friction, and modeling approximations.

References

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- [7] Quanser Inc. SRV02 Rotary Flexible Link User Manual, 2011.
- [8] Norman S. Nise. Control Systems Engineering. John Wiley & Sons, Inc., 2008.
- [9] Quanser Inc. Q2-USB User's Manual, 2010.
- [10] Quanser Inc. VoltPAQ User's Manual, 2010.

Appendix A

MATLAB Code:

```
% Author: Anuranan Bharadwaj  
% Lab 3
```

```
figure()  
plot(data_pos(:,1), data_pos(:,2))  
hold on  
plot(data_pos(:,1), data_pos(:,3))  
xlabel("Time (s)")  
ylabel("Load Shaft Position (rad)")  
title("Measured SRV02 PV ramp response")  
legend("Setpoint Position", "Simulated Position")
```